How Fragile Are Pairwise-Comparison Leaderboards? Worst-Case and Benign Perturbations of Bradley–Terry Rankings Across Evaluation Designs

Anonymous Author(s)
Affiliation
Address
email

Abstract

Pairwise-comparison leaderboards such as Chatbot Arena rank participants by a Bradley–Terry (BT) fit to many "A versus B" outcomes and are used for model selection and marketing, yet it is unclear how much the displayed ranking depends on individual comparisons. We measure this with *verified worst-case perturbations*: an influence-function search for the smallest set of comparisons to remove, flip or inject so that a ranking decision changes, with every reported budget confirmed by an exact refit, and compare it with benign noise (bootstrap, random removal or flipping). On four real datasets from three evaluation designs (Arena crowd votes, MT-Bench expert and GPT-4 verdicts, NBA games), the point ranking is often extremely fragile: changing the top-1 requires altering 1–2 of 57 477 Arena comparisons and about 9 of 59 008 NBA games, but 46–166 of 2 400–3 355 MT-Bench verdicts. The budget tracks the target's gap in standard errors (Spearman correlation 0.86–0.96 on the 63 Arena and 44 NBA adjacent pairs) and, in labelled synthetic calibration, scales like $z\sqrt{N/p}$. For Arena and NBA these swaps lie within sampling noise: bootstrap resampling alone changes the Arena top-1 in about half of the resamples, and the cheapest point-rank top-1 attack leaves the confidence-interval-based published rank of Arena and NBA unchanged (we did not search for attacks aimed at the published rank). Differences between datasets are confounded with size, number of participants and gap structure; at matched size and participant count the spread of top-1 budgets shrinks from about 80-fold to about 7-fold, and GPT-4 verdicts on the same questions are more stable than expert votes. Our budgets are upper bounds on the true minimum, and our conclusions concern fixed snapshots and BT fits.

1 Introduction

Leaderboards built from pairwise comparisons—Chatbot Arena being the prominent example [Chiang et al., 2024]—fit a Bradley–Terry (BT) model [Bradley and Terry, 1952] to many "A versus B" outcomes and publish the ranking, often with confidence intervals. Practitioners use such rankings to pick models, and providers cite them in announcements. How much a ranking depends on individual comparisons is therefore a practical question, whether the change comes from noise or from deliberate action [Min et al., 2025, Singh et al., 2025].

We ask: *what is the smallest change to the comparison data that changes a decision a reader takes from the leaderboard, how does this compare with benign noise, and does it differ across datasets and evaluation designs?* We make this operational by (i) fixing decisions (rank 1, the order of adjacent pairs, top-k sets, Kendall's τ, and the published rank derived from confidence intervals), (ii) searching

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for the smallest verified removal, flip or injection that changes each decision, (iii) comparing with bootstrap and random perturbation, and (iv) comparing datasets on a common footing using the standardised gap and size/participant controls instead of raw counts.

Contributions. (1) A verified-budget methodology for rank decisions of BT leaderboards that adapts influence-function robustness analysis [Koh and Liang, 2017, Broderick et al., 2020] and reports approximation errors. (2) Measurements on four real datasets from three evaluation designs with bootstrap spread: point-rank decisions can flip with a handful of comparisons where gaps are small, but on Arena and NBA these flips are within sampling noise, and the cheapest point-rank attack does not change a confidence-interval-based rank (attacks aimed at that rank were not searched). (3) A scale that explains the differences: the budget follows the gap in standard errors, and in labelled synthetic worlds scales like $z\sqrt{N/p}$, so cross-dataset differences track size, participants and gap structure and cannot be attributed to the evaluation design with these data. (4) Explicit scope: our budgets are upper bounds, from fixed snapshots, with optimistic bootstrap intervals.

2 Related Work

Pairwise-comparison leaderboards. The Bradley–Terry (BT) model [Bradley and Terry, 1952] underlies the rankings of Chatbot Arena, which aggregates crowd votes between anonymous model pairs and reports BT scores with confidence intervals [Chiang et al., 2024]. MT-Bench releases expert and GPT-4 pairwise judgments [Zheng et al., 2023], a second, differently produced comparison design. BT estimation has many extensions, among them ties and covariates such as a home-advantage term [Hunter, 2004, Turner and Firth, 2012], which we use for sports data.

Critiques and manipulation of Arena. Min et al. [2025] study vote-rigging strategies that raise a model's rank, and Singh et al. [2025] document data-access and sampling asymmetries in Arena's practice. Both concern *how* the comparison data can be altered or are non-uniformly collected; we ask how much alteration a fixed snapshot tolerates, under removals, flips and injections, and compare this with benign noise and across designs.

Robustness to dropping data. Influence functions [Koh and Liang, 2017] give first-order effects of single observations, and Broderick et al. [2020] turn them into a search for the small subset whose removal most changes an analysis conclusion, verified by refitting. We adapt this approach to ranking decisions of a BT fit.

3 Method

Leaderboard. A dataset is a list of N comparisons $c = (i_c, j_c, y_c)$ between participants $i_c, j_c \in \{1, \dots, p\}$ with outcome $y_c \in \{0, \frac{1}{2}, 1\}$ (score of i_c; $\frac{1}{2}$ is a tie). The leaderboard is the Bradley–Terry (BT) fit [Bradley and Terry, 1952]: $P(i \text{ beats } j) = \sigma(\theta_i - \theta_j)$, and $\hat{\theta}$ minimises the logistic loss $-\sum_c [\, y_c \log \sigma(\eta_c) + (1-y_c)\log(1-\sigma(\eta_c))\,] + \frac{\lambda}{2}\|\theta\|^2$, $\eta_c = \theta_{i_c} - \theta_{j_c}$, with a weak ridge $\lambda = 10^{-3}$ that fixes the additive gauge and keeps the fit finite under perfect separation. Ties count as half a win to each side. For sports data an additional shared home-advantage term enters η_c. The ranking is $\mathrm{argsort}(-\hat{\theta})$; scores are in natural log-odds units. Bootstrap confidence intervals resample the N comparisons with replacement.

Decisions. We do not perturb scores but the *decisions* a reader takes from a leaderboard: (a) the identity of rank 1; (b) the order of an adjacent pair; (c) the top-k set (Appendix Table 5); (d) Kendall's τ against the original ranking falling below 0.95 or 0.9; (e) the *published-style* rank $1 + \#\{j : \mathrm{lo}_j > \mathrm{hi}_i\}$ computed from 95% score intervals, which is what a leaderboard with confidence intervals displays. Targets (the top pair, and every adjacent pair of the original ranking) are fixed by the original ranking before any attack is run.

Perturbation models and budget. We consider three alterations of the comparison data: *drop* m comparisons, *flip* the outcome of m comparisons, and *inject* m new comparisons that involve the target (win for the promoted participant against some opponent, or loss of the leader). The budget is m, also reported as m/N and relative to the number of comparisons carried by the target participants.

Worst-case search. For a target pair (a, b) with a ranked above b we look for a small set of alterations after which $\hat{\theta}_a \leq \hat{\theta}_b$. Per comparison, the first-order effect on $\hat{\theta}_a - \hat{\theta}_b$ is given by the influence function of the penalised BT likelihood [Koh and Liang, 2017], the same approximation that Broderick et al. [2020] use to find the small data subset whose removal changes a conclusion. We rank comparisons by this effect, alter the best batch, refit exactly, and repeat; then we bisect over the prefix of the alteration order, so the reported budget $m^\star$ is always verified by an exact refit of the altered data. For injection, all m new votes involve the target and a single opponent, chosen among the six most promising by the first-order screen and verified by exact refits. Attacks that succeed only by disconnecting the comparison graph are discarded. Because the search is greedy, $m^\star$ is an *upper bound on the minimum* number of alterations: a found attack proves that the decision is that fragile, while the absence of an attack proves nothing about robustness; we have no certified lower bound. For Kendall's τ we apply the cheapest adjacent-swap attack repeatedly until τ drops below the threshold. Baselines restrict the alterations to head-to-head comparisons of the target pair, or to comparisons involving either target.

Benign noise and scale. As benign perturbations we use bootstrap resampling, random dropping and random flipping of m comparisons; we report the probability that each decision changes (Wilson intervals). To compare datasets of different size we express the scale of a decision in two ways: the gap of the targeted pair in bootstrap standard-error units, and the subsample fraction b_q of N at which the decision flips with probability q under random subsampling (reported with censoring flags when outside the grid).

4 Experimental Setup

Data. We use four real comparison sets from three evaluation designs, all public snapshots pinned by commit and checksum (Appendix B). *Crowd votes:* Chatbot Arena human preference data (57 477 battles between 64 models; HF `lmarena-ai/arena-human-preference-55k`, apache-2.0). Ties and "both bad" are merged into ties. *Expert and model-judge verdicts:* MT-Bench human judgments (3 355 votes on 6 models) and GPT-4 pairwise judgments of the same models (2 400 verdicts; "inconsistent" verdicts are treated as ties) [Zheng et al., 2023]; 1 814 questions are covered by both. *Sports:* the FiveThirtyEight NBA game file, one row per game after removing the duplicate `_iscopy` rows, NBA league only (the 1968–76 ABA is a separate, weakly connected league and is analysed as its own leaderboard where mentioned), 59 008 games between 45 franchises when pooled over all seasons and 78 single regular seasons analysed separately; the BT model has one shared home-advantage parameter. We did not use the optional larger Arena set, WebDev Arena, or the LLM-judge verdicts on Arena battles (deferred in the study plan for budget reasons).

Targets, seeds and repeats. Targets are fixed from the original ranking before attacks run: the top-1 pair, every adjacent pair, and Kendall's τ thresholds. Bootstrap resamples (with replacement over comparisons) quantify the sampling spread of each adversarial budget; random-perturbation baselines use Monte-Carlo draws; all seeds are fixed in the scripts (Appendix B). Arena rows have no user identifier and repeat prompts (51 734 distinct prompts in 57 477 rows), so treating comparisons as independent makes bootstrap intervals optimistic.

Controls for size and design. Datasets differ in size, number of participants, tie rate and structure, so cross-dataset differences are not attributed to "design". Clean contrasts are (i) MT-Bench expert versus GPT-4 verdicts on the same questions, (ii) Arena subsampled to the size of MT-Bench (all 64 models) and restricted to its six strongest models, and (iii) variation across NBA seasons.

Prior work on this question. The task statement noted that its question has been studied in a published paper that must not be consulted. One starter-list entry appeared to match that description, so we did not read, cite or search for it; no forbidden material was encountered (see `REPRODUCE.md`).

Compute and LLM assistance. All experiments run on CPU (2 cores). LLM assistance (Claude Sonnet 5.5, via Claude Code) was used for literature reading, code, analysis and writing of this paper and all numbers, figures and tables are produced by the shipped code.

5 Results

All budgets are the smallest verified attacks we found (upper bounds on the minimum). "Remove", "flip" and "inject" are the three alteration types; N is the number of comparisons.

Point rankings can flip with a handful of comparisons. Table 1 gives the smallest attack changing the top-1 and pushing Kendall's τ below 0.95. On Arena-55k (57 477 battles, 64 models) the top-1 changes after removing 2 comparisons, flipping 1, or injecting 2 (0.003% of N); in the pooled NBA data (59 008 games) after removing 9 or flipping 6. On MT-Bench the same decision needs 92 removals or 46 flips with expert votes ($N = 3355$) and 166 or 61 with GPT-4 verdicts ($N = 2400$), i.e. 1.4–6.9% of N. The ranking as a whole is less brittle than its leader: for $\tau < 0.95$ the smallest attacks remove 56 comparisons on Arena (0.10%), 21 on NBA (0.04%), 13 on MT-Bench expert (0.39%) and 82 on GPT-4 (3.4%). In one of six resamples the Arena $\tau < 0.9$ flip attack found no attack within 50% of N (the budget there is censored); for pair decisions the subsample fraction b_{50} is not reached, so we report b_{10} and b_{25} with grid flags in `results/summary_b50.csv`. Over all adjacent pairs of the original ranking, 95% of the Arena pairs and 95% of the NBA pairs have a gap below two standard errors, 63% and 70% respectively can be swapped by at most ten removals, and the median swap needs 5 (Arena) or 5.5 (NBA) removals. The smallest swap budget rises with the gap z (Figure 1, Appendix Table 6; Spearman correlation between budget and z: 0.96 and 0.95 on Arena (63 pairs; removal, flip), 0.87 and 0.86 on NBA (44 pairs), 0.97–1.0 on the five MT-Bench pairs). Over 78 single NBA regular seasons the top-1 changes after a median of 4 removals (95% bootstrap interval of the median 3–5; 0.56% of the season's games) at a median leader gap of 0.64 standard errors. Of the 78 season leaderboards, 12 have an exact tie between the top two teams (budget 0). The first-order influence estimate matches the exact-refit budget on Arena and NBA (median ratio 1.0) but is off by a factor 0.94–1.97 on MT-Bench, where budgets are large; the reported budgets are always the verified ones.

On Arena and NBA these changes are within sampling noise. Bootstrap resampling of all comparisons changes the Arena top-1 in 48% of 100 resamples (Wilson 95% interval 38–58%) and the NBA top-1 in 37% (28–47%), but never in 200 benign bootstrap resamples of either MT-Bench source (where the budgets are large) (one-sided 95% bound 1.9%). Random removal or flipping of as many comparisons as the cheapest attack changes the top-1 only rarely (Table 3, Figure 3): for example 0 of 200 random removals of 2 comparisons and 4 of 200 random flips of 10 on Arena. Sampling-scale noise is not negligible: even when random subsamples keep 90% of the data the Arena top-1 differs from the full-data leader in 46% of 100 draws, and in 24% for NBA (`results/subsample_*.csv`). Random flips do move the Arena top-1 once they reach roughly 0.1–1% of N (Figure 3b). So the worst case needs far fewer alterations than random noise of the same size would use to move the ranking, but it exploits a gap that sampling already leaves open. The published-style rank, which accounts for confidence intervals, is unaffected: after the cheapest top-1 attack on Arena or NBA no model changes its published rank, although two models change their point rank; on MT-Bench one or two models change their published rank at the larger budgets (Table 2).

Design versus size. Across datasets the top-1 budget differs by about 80-fold in comparison counts (2 on Arena, 166 on MT-Bench GPT-4; Table 1), but datasets differ in size, number of participants, tie rate and gap structure at once. Figure 2(b) separates some factors. Subsampling Arena to the size of MT-Bench ($N = 2400$ or 3355, all 64 models) leaves the top-1 budget at 1–2 removals (median over 50 subsamples), as on the full data, so size alone does not explain the MT-Bench robustness. Restricting Arena to its six highest-ranked models, which are near-tied GPT-4 variants, raises it to a median of 11.5 removals on 2 400 comparisons (14 on all 3 222); this control is not comparable with the random-six control below. In the control with $N = 2000$ comparisons and six participants (six random models or teams among the 20 most compared; for MT-Bench the six fixed models), the median top-1 removal budget is 60 on Arena, 20.5 on NBA, 56 on MT-Bench expert and 138.5 on MT-Bench GPT-4. The spread is then about 7-fold instead of 80-fold, with the GPT-4 verdicts the most stable. With $N = 2000$ and all participants the median budget is 2 on Arena (the graph was connected in 99 of 100 subsamples) and 55 and 140.5 on MT-Bench; the same control on NBA disconnects the graph in most subsamples and is not used. On the same 1 814 questions (Figure 2(a)), the removal budget for the top-1 is 103.5 (bootstrap interval of the median 96–110) for GPT-4 verdicts, but 41 (39–43) for one randomly chosen expert vote per question; the standardised top-1 gap on the

Table 1: Smallest verified attack found, in percent of N (bootstrap 2.5–97.5 percentiles in brackets, from 20 resamples for the Arena and NBA top-1 rows and 100 for MT-Bench; the Arena and NBA τ rows have only 6 resamples and show the point value). Top-1: the decision changes; τ: Kendall's τ against the original ranking falls below 0.95.

Dataset	N	p	remove	flip	inject
top-1 changes					
Arena 55k	57 477	64	0.003 [0.003, 0.091]	0.002 [0.002, 0.059]	0.003 [0.003, 0.12]
MT-Bench expert	3 355	6	2.74 [1.58, 4.13]	1.37 [0.80, 1.97]	2.71 [1.58, 3.92]
MT-Bench GPT-4	2 400	6	6.92 [4.25, 8.86]	2.54 [2.02, 2.96]	5.04 [4.02, 5.92]
NBA	59 008	45	0.015 [0.002, 0.070]	0.010 [0.002, 0.045]	0.019 [0.003, 0.088]
Kendall $\tau < 0.95$					
Arena 55k	57 477	64	0.097	0.061	–
MT-Bench expert	3 355	6	0.39 [0.030, 1.54]	0.21 [0.030, 0.77]	–
MT-Bench GPT-4	2 400	6	3.42 [2.23, 4.25]	1.62 [1.12, 1.96]	–
NBA	59 008	45	0.036	0.063	–

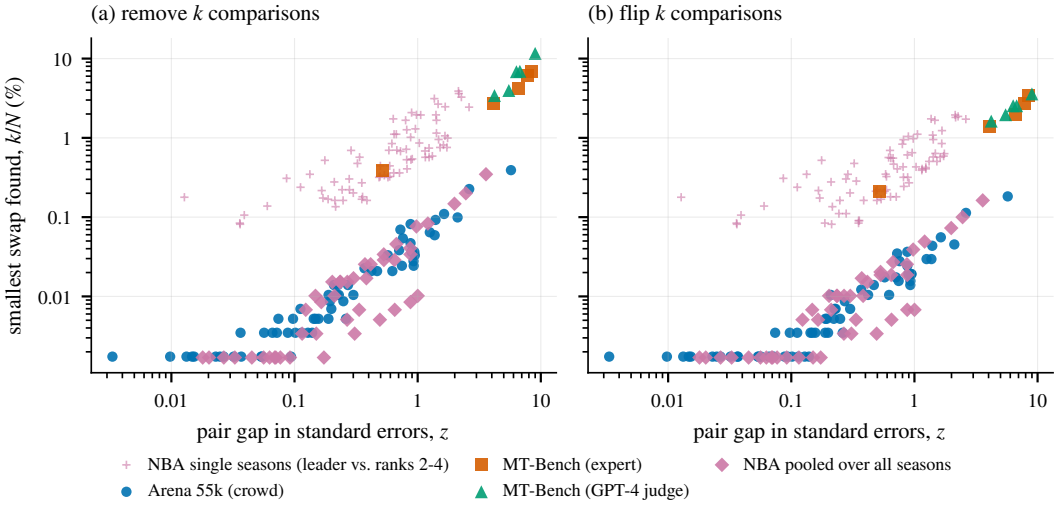


Figure 1: Smallest verified swap of an adjacent pair, as a fraction of N, against the pair's gap in standard errors; (a) removals, (b) flips; log-log axes. Markers: adjacent pairs of the full-data ranking of each dataset (NBA pooled over all seasons); crosses: NBA single seasons (leader against ranks 2–4). Many Arena pairs sit at the floor of one comparison. Upper bounds on the minimum.

original data is 5.5 for GPT-4 and 4.1 for all expert votes. This is a contrast of two judges on the same items, not of "experts versus crowds".

Modelling choices. The Arena leader is itself sensitive to the tie rule: with ties dropped (39 716 battles) the leader changes from gpt-4-1106-preview to gpt-4-0125-preview, with the top-1 removal budget rising from 2 to 9 (Appendix Table 4). Ridge strength (from 10^{-5} to 10^{-1}) and, for NBA, the home term change the ranking little (Kendall $\tau \geq 0.976$ with the primary fit) and leave the top-1 budgets within 1 comparison.

What predicts fragility. In simulated BT worlds (**synthetic**; calibration only) the log top-1 removal budget regresses on $\log z$ (coefficient 0.91, 95% interval 0.85–0.96), $\log N$ (0.49) and $\log p$ (-0.49) with $R^2 = 0.98$, i.e. the budget scales roughly like $z\sqrt{N/p}$ (Appendix Figure 4). For the cheapest adjacent swap the fitted coefficient on $\log p$ is -1.93 (95% interval -2.12 to -1.76) and on $\log z$ 0.08 (`results/summary_synthetic_regression.csv`): with many participants some pair is nearly tied, so the first swap is cheap.

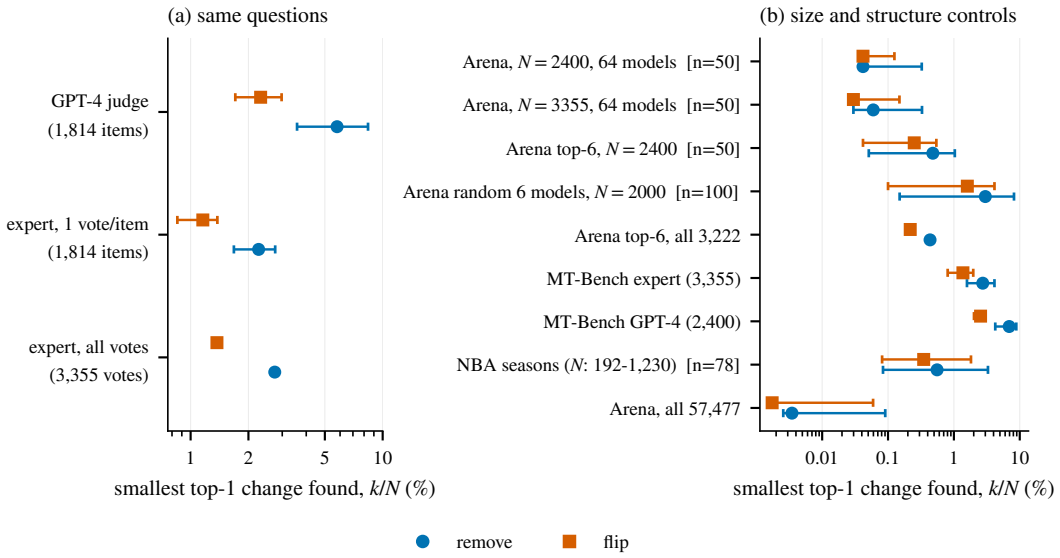


Figure 2: Top-1 budget. (a) The same 1 814 MT-Bench questions judged by experts and by GPT-4. (b) Size and structure controls fixed before running; rows marked $[n{=}\cdot]$ show the median over n subsamples with 2.5–97.5 percentile bars; full-data rows show the original-data value with a 2.5–97.5 percentile bootstrap bar where resamples exist; "1 vote/item" is the median over 100 random one-vote-per-question draws. The top-6 rows (the six highest-ranked models, near-tied GPT-4 variants) are not comparable with the random-six-models row. Cross-dataset differences are confounded with N, number of participants, ties and structure.

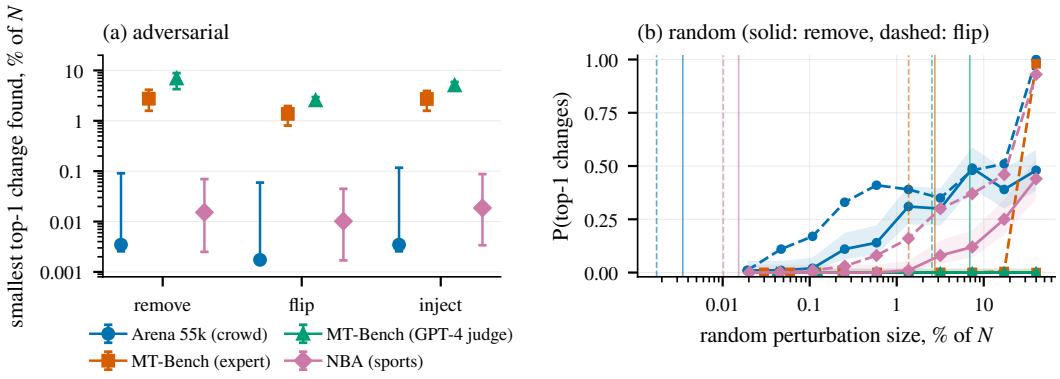


Figure 3: (a) Smallest verified top-1 attack by removing, label-flipping or injecting votes: dot, original data; bar, 2.5–97.5 percentile over bootstrap resamples (drawn for at least 20 resamples; Arena and NBA have 20). (b) Probability that random removal (solid) or flipping (dashed) of m comparisons changes the top-1, colours as in (a); Wilson 95% band for removal only; vertical lines mark the adversarial budgets.

6 Discussion and Limitations

What the evidence says. Fragility of a *point* ranking is large where the ranking has near-ties, and a verified attack can be tiny: a few comparisons out of tens of thousands. But on Arena and NBA this size is of the order of sampling noise; resampling the same data already changes the Arena top-1 about half the time, and the confidence-interval-based rank was not moved by the cheapest top-1 attack on Arena or NBA. Reading the point order as a fact is therefore the fragile step; the intervals were not moved by the point-rank attack we ran, but we did not search for attacks aimed at the published rank, so we do not claim that it is robust. At the other extreme, MT-Bench's six clearly separated models need dozens to hundreds of comparisons (1.4–6.9% of N) to change the top-1.

Table 2: Effect of the cheapest top-1 attack (removal or flip, k comparisons) on the published-style rank (95% Wald intervals): the new point-rank leader, the number of models at published rank 1 before $\rightarrow$ after, and the number of models whose point rank (Δ point) or published rank (Δ pub.) changes. Published rank 1 is shared when intervals overlap.

Dataset	attack	k	new leader	# at pub. rank 1	Δ point ranks	Δ pub. ranks
Arena 55k	drop	2	gpt-4-0125-preview	$2 \rightarrow 2$	2	0
Arena 55k	flip	1	gpt-4-0125-preview	$2 \rightarrow 2$	2	0
MT-Bench expert	drop	92	claude-v1	$1 \rightarrow 2$	2	2
MT-Bench expert	flip	46	claude-v1	$1 \rightarrow 2$	2	2
MT-Bench GPT-4	drop	166	claude-v1	$1 \rightarrow 2$	2	1
MT-Bench GPT-4	flip	61	claude-v1	$1 \rightarrow 2$	2	1
NBA	drop	9	Lakers	$4 \rightarrow 4$	2	0
NBA	flip	6	Lakers	$4 \rightarrow 4$	2	0

Table 3: Smallest top-1 attack (k comparisons) against random removal or flip of $m = k$ and $m = 10k$ comparisons: probability that the top-1 changes (Wilson 95% interval, 200 draws; 500 for MT-Bench expert).

	remove			flip		
Dataset	attack k	random, $m{=}k$	random, $m{=}10k$	attack k	random, $m{=}k$	random, $m{=}10k$
Arena 55k	2	0.00 [0.00, 0.02]	0.00 [0.00, 0.02]	1	0.01 [0.00, 0.04]	0.02 [0.01, 0.05]
MT-Bench expert	92	0.00 [0.00, 0.01]	0.00 [0.00, 0.01]	46	0.00 [0.00, 0.01]	0.00 [0.00, 0.01]
MT-Bench GPT-4	166	0.00 [0.00, 0.02]	0.00 [0.00, 0.02]	61	0.00 [0.00, 0.02]	0.01 [0.01, 0.04]
NBA	9	0.00 [0.00, 0.02]	0.00 [0.00, 0.02]	6	0.00 [0.00, 0.02]	0.00 [0.00, 0.02]

Cross-dataset differences track how many participants share how many comparisons and how far apart their scores are; with one dataset per design they cannot be attributed to the evaluation design. At matched N and participant count a 7-fold spread remains (20.5 to 138.5 comparisons), and the two judges on the same questions differ by about 2.5-fold. The one design contrast we can make cleanly (experts versus GPT-4 verdicts on the same questions) favours the model judge as the more stable ranking, which says nothing about its validity.

Limitations. (i) Budgets are *upper bounds on the minimum*: the greedy influence search can miss smaller attacks, we have no certified lower bound, and a failure to find an attack is not evidence of robustness. (ii) Bootstrap intervals treat comparisons as independent; Arena rows carry no user identifier and repeat prompts, so repeated votes by one user or on one prompt would make them too narrow. Arena and NBA attack bootstraps use 20 resamples (6 for the τ attacks), so those spreads are coarse. (iii) We analyse fixed snapshots (Arena-55k covers 2023–24) with one BT specification (ties as half wins, weak ridge, shared home term); Davidson and Rao–Kupper tie models were not run, though sensitivity to ties, ridge and home term is in the appendix tables. The live leaderboards use more data and their own conventions, which we do not reproduce. (iv) Our perturbations are idealised: injected votes are assumed to be accepted, and removed or flipped votes are chosen with full knowledge of the data, which a real adversary does not have. We did not model Arena's sampling of model pairs or the private testing described by Singh et al. [2025]. (v) Design effects are confounded with data size and structure except in the MT-Bench contrast; each design has essentially one dataset (NBA seasons give only within-design variation). Judge-model verdicts on Arena battles, Arena-140k and WebDev Arena were not analysed. (vi) MT-Bench human votes include several judges per question, and the all-history NBA fit is misspecified over decades (seasons are the sensible unit); online Elo, Davidson and Rao–Kupper tie models, Arena model-name merging, and the optional datasets above were not run. (vii) Synthetic worlds are only a calibration of the scaling law; we make no claim about real leaderboards from them.

Broader impact. The method can be read as a recipe for manipulating leaderboards; it requires access to the comparison data, which is public only for snapshots, and we think the main use is auditing: reporting rank intervals, not point ranks, and the number of comparisons separating neighbours.

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A Modelling-choice sensitivity and synthetic calibration

Table 4 refits the leaderboards with ties dropped, a ridge of 10^{-1} or 10^{-5} and, for NBA, without the home term; the ridge and the home term barely matter, while dropping ties changes the Arena leader. Figure 4 shows the **synthetic** calibration worlds (simulated Bradley–Terry data with known scores, used only to check how the budget scales with the gap, N and p, never as a result on real leaderboards).

Table 4: Modelling variants: Kendall τ of the variant's ranking with the primary one, whether the leader is the same as under the primary fit, the standardised top-1 gap, and the smallest verified removal and flip budgets for the top-1.

Dataset	variant	N	τ vs. primary	leader	z_{top}	remove k	flip k
Arena 55k	primary	57477	1.000	same	0.06	2	1
	ties dropped	39716	0.980	changed	0.38	9	5
	ridge 10^{-1}	57477	0.999	same	0.06	2	1
	ridge 10^{-5}	57477	1.000	same	0.06	2	1
MT-Bench expert	primary	3355	1.000	same	4.11	92	46
	ties dropped	2575	1.000	same	4.63	83	42
	ridge 10^{-1}	3355	1.000	same	4.10	92	46
	ridge 10^{-5}	3355	1.000	same	4.11	92	46
MT-Bench GPT-4	primary	2400	1.000	same	6.76	166	61
	ties dropped	1800	1.000	same	6.08	75	39
	ridge 10^{-1}	2400	1.000	same	6.76	166	61
	ridge 10^{-5}	2400	1.000	same	6.76	166	61
NBA	primary	59008	1.000	same	0.23	9	6
	ridge 10^{-1}	59008	1.000	same	0.23	9	6
	ridge 10^{-5}	59008	1.000	same	0.23	9	6
	no home term	59008	0.976	same	0.17	9	5

Table 5: Smallest verified attack changing the top-k set (comparisons altered, original data; 2.5–97.5 bootstrap percentiles where at least 20 resamples exist; "–": top-10 is undefined for the six MT-Bench models). Such sets are the most fragile decision at the boundary, where models are statistically tied (e.g. Arena top-10 and NBA top-5 change after 1 comparison).

		top-3 set		top-5 set		top-10 set	
Dataset	N	remove	flip	remove	flip	remove	flip
Arena 55k	57 477	35 [11, 194]	22 [7, 95]	3 [1, 36]	2 [1, 18]	1 [1, 24]	1 [1, 14]
MT-Bench expert	3 355	206 [168, 252]	89 [75, 115]	141 [109, 181]	67 [52, 88]	–	–
MT-Bench GPT-4	2 400	164 [103, 229]	61 [47, 75]	95 [67, 138]	47 [34, 59]	–	–
NBA	59 008	11 [1, 158]	8 [1, 77]	1 [1, 46]	1 [1, 26]	1 [1, 21]	1 [1, 13]

Table 6: Smallest adjacent-swap budget against the pair's gap z (Figure 1): number of adjacent pairs, Spearman correlation between budget k and z, and slope of $\log k$ on $\log z$. With $n = 5$ pairs the MT-Bench correlations are nearly uninformative.

Dataset	attack	n pairs	Spearman(k, z)	log-log slope
Arena 55k	remove	63	0.96	0.87
Arena 55k	flip	63	0.95	0.74
MT-Bench expert	remove	5	1.00	1.00
MT-Bench expert	flip	5	1.00	0.95
MT-Bench GPT-4	remove	5	1.00	1.70
MT-Bench GPT-4	flip	5	0.97	1.06
NBA	remove	44	0.87	1.01
NBA	flip	44	0.86	0.86

B Reproducibility

Reproduction. Raw data are kept outside the code repository; `code/00_fetch_data.sh` downloads and checks the pinned snapshots: Arena-55k (Hugging Face `lmarena-ai/arena-human-preference-55k`, commit `18c29834`; `train.csv` sha256 `0692154a...f5be2`), MT-Bench human and GPT-4 judgments (`lmsys/mt_bench_human_judgments`, commit `f7d2896d`; human parquet sha256 `4877bc46...5769`), NBA (`fivethirtyeight/data`, `nba-elo/nbaallelo.csv`, commit `4c1ff5e3`; sha256 `d46ed354...16bc`). Licences: Arena-55k apache-2.0; MT-Bench judgments cc-by-4.0; NBA data CC-BY-4.0. All experiments ran on CPU (2 cores, shared); the longest single scripts (Arena and NBA attack bootstraps, common-size controls, synthetic calibration) took 10–20 minutes. Software: Python 3.11.2, numpy 2.4.6, scipy 1.17.1, pandas 3.0.6 (`results/ENV.txt`). Seeds (fixed in the scripts): attack bootstrap resample r uses seed $1000 + r$; benign Monte-Carlo 2024; subsample curves 5; MT-Bench contrast 3; common-size controls 77 and Arena controls 78; synthetic worlds 9; equal-size random baselines 11; bootstrap intervals of summary medians 6. The reproduction guide `REPRODUCE.md` lists environment versions, seeds, commands, run times, and the script that generates each table and figure.

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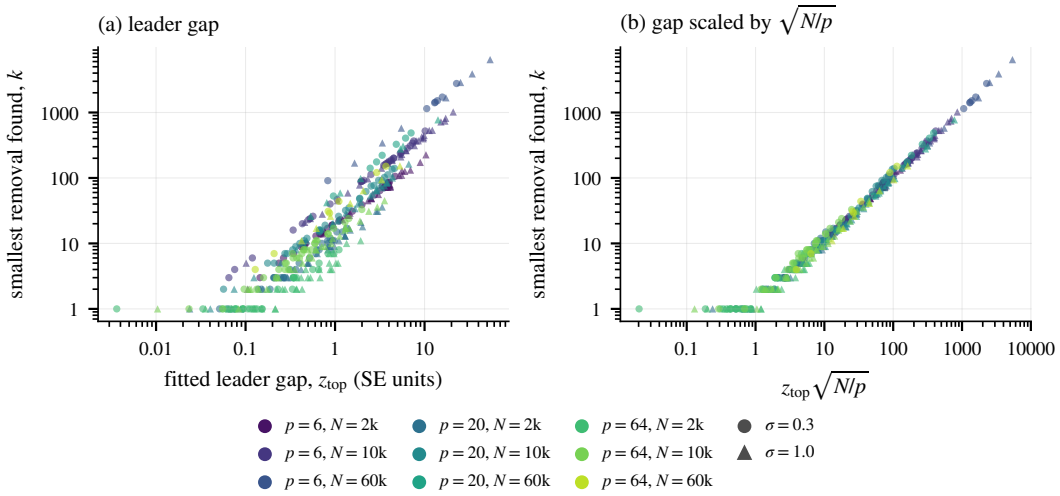


Figure 4: **Synthetic** Bradley–Terry worlds (not real leaderboards): smallest verified top-1 removal budget (a) against the fitted leader gap z_{top}, for several N, p and score spreads σ, and (b) against $z_{\text{top}}\sqrt{N/p}$, where the points collapse onto one line (the regression in `results/summary_synthetic_regression.csv`).

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